

Abhay Patel

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EXPERIENCE

Tata Consultancy Services (TCS)

Aug. 2024 – Present

Data Analyst / Python Developer

Mumbai, India

- Analyzed and validated **high-volume banking transaction data** using **PostgreSQL and SQL queries**, identifying data inconsistencies, transaction patterns, and exceptions across enterprise banking workflows
- Performed **root cause analysis** for production issues by correlating **application logs, API responses, transaction data, and database records**, enabling faster identification and resolution of data-related issues
- Developed **Python-based automation scripts** for recurring data validation, monitoring, and reporting tasks, reducing manual effort and improving operational efficiency
- Performed **data validation, reconciliation, and quality checks** during SIT/UAT cycles by comparing system-generated outputs with expected business outcomes and defined business rules
- Extracted, transformed, and analyzed data using **SQL** to support reporting, investigate anomalies, identify failure patterns, and provide actionable insights to business and technical teams
- Analyzed **REST API requests and responses** to trace transaction journeys, validate data flow between systems, and troubleshoot integration and data discrepancies
- Collaborated with cross-functional teams to investigate business requirements, analyze operational data, troubleshoot data issues, and support **data-driven decision-making** in a high-volume enterprise environment

Prism IT Systems

Jan. 2024 – Jun. 2024

Full Stack Developer (Next.js, TypeScript)

Surat, India

- Developed and maintained **full stack web applications** using **Next.js, React, TypeScript, Node.js, and REST APIs**, translating business requirements into scalable technical solutions
- Designed and integrated **REST APIs** for data exchange between frontend applications and backend services, implementing validation, authentication, and error handling
- Worked with **MongoDB** to design data models, perform CRUD operations, and structure application data for efficient retrieval and reporting
- Analyzed application workflows and database records to identify functional issues, troubleshoot data inconsistencies, and improve application reliability
- Used **Git, Postman, and debugging tools** for API testing, version control, issue investigation, and development workflow management

PROJECTS

AI-Powered Customer Churn Analysis & Prediction Platform

- Built an end-to-end **customer churn analysis and prediction platform** using **Python, SQL, Pandas, NumPy, and Scikit-learn** to identify high-risk customers and predict churn probability
- Performed **data cleaning, exploratory data analysis (EDA), feature engineering, and statistical analysis** to identify customer behavior patterns and key churn-driving factors
- Developed and evaluated **classification models** using metrics including accuracy, precision, recall, F1-score, and ROC-AUC to select an appropriate predictive model
- Created **Power BI dashboards** to visualize churn trends, customer segments, risk distribution, and business KPIs for data-driven decision-making
- Automated data preprocessing and reporting workflows using **Python**, reducing repetitive manual analysis and improving reporting efficiency

SKILLS

Programming Languages : Python, SQL

Big Data Technologies : PySpark, Apache Spark

Data Analysis Libraries : Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Databases : PostgreSQL, MySQL

Data Analysis : Data Cleaning, Exploratory Data Analysis (EDA), Data Validation, Data Transformation, Feature Engineering, Statistical Analysis, Root Cause Analysis, Trend Analysis, KPI Monitoring

Visualization & Reporting : Power BI, Excel, Pivot Tables, VLOOKUP/XLOOKUP, Power Query, Interactive Dashboards, Data Storytelling

AI & Automation : OpenAI APIs, LangChain, LangGraph, CrewAI, AI Agents, Workflow Automation, Python Scripting

Tools : Jupyter Notebook, Git, GitHub, VS Code, Postman, Jira

Concepts : ETL Pipelines, Data Warehousing, Banking Domain Analytics, Business Analytics, SDLC, Agile Methodology

EDUCATION

LD College Of Engineering, Ahmedabad, Gujarat

Aug. 2020 – May 2024

Bachelor of Engineering [ECE]

CGPA: 8.03/10.00

- Coursework**: DBMS, OOPS, Computer Networks, Machine Learning